Yu Yuan

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Research Interests

Physics-Consistent and Controllable Video Generation, Physical World Models, Dynamics Modeling, 3D/4D Generation, Computational Photography, Image/Video Editing, Image/Video Enhancement.

Education

Purdue University — West Lafayette, IN, US
PhD Student in Electrical and Computer Engineering Advisor: Stanley H. Chan — *08/2023 – 05/2027 (expected)*

Shanghai Jiao Tong University — Shanghai, China
MS in Aeronautical and Astronautical Science and Technology — *09/2020 – 06/2023*

Shanghai Jiao Tong University — Shanghai, China
BEng in Aeronautics and Astronautics Engineering — *09/2016 – 06/2020*
Minor Degree in Administration Management — *10/2018 – 06/2020*

Projects Experience

Adobe Research — San Jose, CA
Research Intern (Video Foundation Model Team) — *05/2026 – Present*

- **Conducting** research on video foundation models, controllable video generation, and generative video editing.
- **Researching** controllable and physics-aware generative video models for video synthesis, editing, and world dynamics simulation.

Purdue University — West Lafayette, IN
Research Assistant (Advisor: Prof. Stanley H. Chan) — *08/2023 – Present*

- **OptiWorld (Optimal-Controlled Video World Generation)**:
  - ⋆ **Introduced** optimal control into video generation by optimizing object dynamics under physical constraints.
  - ⋆ **Developed** a continuous manifold optimization framework that transforms safety, efficiency, and smoothness objectives into unified trajectory planning for controllable world generation.
- **SeeU (4D Understanding and Video Generation)**:
  - ⋆ **Proposed** SeeU, a unified framework for continuous 4D spatiotemporal world understanding and generation.
  - ⋆ **Pioneered** learning continuous 4D dynamics from 2D inputs to "see the unseen," enabling generation at unseen time and space as well as controllable video editing.
- **NewtonGen (Physics-Consistent Text-to-Video Generation)**:
  - ⋆ **Developed** a physics-consistent and controllable text-to-video framework that explicitly incorporates learnable dynamics.
  - ⋆ **Innovated** Neural Newtonian Dynamics, which models different dynamics via unified neural ODEs.
- **Generative Photography (Scene-Consistent Camera-Controlled Text-to-Image Generation)**:
  - ⋆ **Introduced** a new text-to-image generation paradigm for photography with an understanding of camera physics.
  - ⋆ **Produced** significantly more scene-consistent photorealistic images than SOTA models (Stable Diffusion 3, FLUX).
- **High Dynamic Range (HDR) Imaging**:
  - ⋆ **Proposed** a novel high dynamic range (HDR) imaging framework capable of handling a flexible number of causal inputs.
  - ⋆ **Utilized** side information and attention mechanisms to effectively suppress artifacts during the fusion process.

Selected Publications

Yu Yuan, Jianhao Yuan, Xijun Wang, Daiqing Li, Liu He, Lu Ling, Stanley H. Chan.
OptiWorld: OptiWorld: Optimal Control for Video World Generation under Physical Constraints
2026 [Under Review]

Yu Yuan, Tharindu Wickremasinghe, Zeeshan Nadir, Xijun Wang, Yiheng Chi, Stanley H. Chan.
SeeU: Seeing the Unseen World via 4D Dynamics-aware Generation
CVPR 2026

Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, Bole Ma, Stanley H. Chan.
NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics.
ICLR 2026

Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, Stanley H. Chan.
Generative Photography: Scene-Consistent Camera Control for Realistic Text-to-Image Synthesis.
CVPR 2025 [Highlight & Demo]

Yu Yuan, Yiheng Chi, Xingguang Zhang, Stanley H. Chan.
iHDR: Iterative HDR Imaging with Arbitrary Number of Exposures.
ICIP 2025

Xingguang Zhang, Nicholas Chimitt, Xijun Wang, **Yu Yuan**, Stanley H. Chan.
Learning Phase Distortion with Selective State Space Models for Video Turbulence Mitigation.
CVPR 2025 [Highlight]

Lanqing Guo*, Xijun Wang*, Minchul Kim*, **Yu Yuan**, Wes Robbins, Xingguang Zhang, Stanley H. Chan, Zhangyang Wang, Xiaoming Liu. ***Beyond Clean Pixels: Interdisciplinary Lessons from Turbulent Long-Range Face Recognition.***
2025 [Under Review]

Joonyeoup Kim, **Yu Yuan**, Xingguang Zhang, Xijun Wang, Stanley H. Chan.
Astrophotography Turbulence Mitigation via Generative Models.
ICIP 2025

Lindong Wang, Hongya Tuo, **Yu Yuan**, Henry Leung, Zhongliang Jing.
RCMixer: Radar-Camera Fusion Based on Vision Transformer for Robust Object Detection.
JVCI 2024

Yu Yuan, Jiaqi Wu, Lindong Wang, Zhongliang Jing, Henry Leung, Shuyuan Zhu, Han Pan.
Learning to Kindle the Starlight.
arXiv 2022

Yu Yuan, Jiaqi Wu, Zhongliang Jing, Henry Leung, Han Pan.
Multimodal Image Fusion based on Hybrid CNN-Transformer and Non-local Cross-modal Attention.
arXiv 2022

Patent

A trajectory solving and alignment method based on inertial measurement unit and ultra-short baseline positioning sensors for autonomous underwater robots. China Patent. CN 114993313 B.

Awards & Competitions

Ivy League Scholarship for Exceptional Students (Innovation)	Top 1%
Outstanding Graduate of Shanghai Jiao Tong University	Top 15%
Shanghai International Creators Competition: Special Competition of Drones	First Prize
DELL AI for Social Innovation Competition: AI/ADAS of Intelligent Car	National Second Place

Other Activities

Teaching Assistant, Automatic Control Theory, Shanghai Jiao Tong University 2021

Workshop Organizer, WACV GAIP 2026

Reviewer, CVPR, ICCV, ICLR, NeurIPS, WACV, CVPRW, ICIP, ICASSP

IEEE Graduate Student Member

Skills

Languages: English (Fluent), Chinese (Native)

Programming: Python, PyTorch, HTML, JavaScript, Matlab, C/C++ (Basic)

Technologies: Diffusers, Transformers, Post-training, 3DGS, CUDA, OpenCV, NumPy, Git, LaTeX, Linux, Slurm, Markdown, Gradio, NVIDIA Jetson, Adobe Photoshop, Adobe Premiere

Others: Photography, Fishing, Drones, Rowing